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Research paper

Heterogeneous trajectories of adolescent depression and risk of suicidal ideation in emerging adulthood: A seven-wave latent class growth analysis of the Korean Children and Youth Panel Survey 2018

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ABSTRACT

Background: Adolescent suicide remains a major public health concern in Korea, yet depressive trajectories from age 13 to emerging adulthood have rarely been examined in nationally representative seven-wave panel data.
Methods: In the Korean Children and Youth Panel Survey 2018 middle-school first-grade cohort ($N = 2452$; age 13 at Wave 1; 80% retained at Wave 7), latent class growth analysis (LCGA) was fitted on a 9-item depression scale excluding the suicidal-ideation item and replicated in R lcmm and Mplus. Distal suicidal ideation (SI) used broad SI (E04 $\geq$ 2) and affirmative SI endorsement (E04 $\geq$ 3), with missing SI retained as missing. Distal outcomes used Bolck-Croon-Hagenaars (BCH) correction; time-to-event analysis used a class $\times$ wave Cox model with ID-level bootstrap (B = 1000).
Results: A four-class solution balanced fit, adjacent-class tests, and interpretability (BIC = 25,228; entropy = 0.66): Stable-Low (25.1%), Persistent-High (10.0%), Decreasing (29.6%), and Late-Increasing (35.3%). BCH-corrected W7 broad SI prevalence was 9.4%, 56.7%, 36.8%, and 66.2%, respectively. Odds ratios versus Stable-Low were 12.63, 5.61, and 18.86 for Persistent-High, Decreasing, and Late-Increasing. The Late-Increasing versus Persistent-High contrast was borderline (OR = 1.49, 95% CI = 1.00–2.25, $p = .052$), indicating two high-risk pathways rather than a definitive hierarchy.
Limitations: SI was measured with a single self-report item; entropy was moderate; and trajectory classes overlap calendar time with repeated SI outcomes.
Conclusions: Persistent-High and Late-Increasing represented two high-risk pathways differing mainly in timing. Repeated assessment across adolescence may help detect late-emerging SI risk.

1. Introduction

Suicide has remained the leading cause of death among Korean adolescents and emerging adults for more than a decade (Statistics Korea, 2024), with substantial 12-month suicidal ideation (SI) prevalence in nationally representative surveys (Kim et al., 2023; Park and Lee, 2024). Major depressive symptoms are the most consistently identified clinical correlate (World Health Organization, 2023), but which adolescents to target remains contested: cross-sectional risk profiling at any single age misses the intra-individual variation in symptom course that distinguishes transient from rising or chronic distress.

Latent class growth analysis (LCGA; Nagin, 1999, 2005) and related finite-mixture growth models are now standard tools for characterising this heterogeneity. Western adolescent samples typically yield three to five trajectories: a stable-low/resilient class, a chronic class, and one or two intermediate classes (Stoolmiller et al., 2005; Costello et al., 2008; Yaroslavsky et al., 2013; Dekker et al., 2007). Class proportions, predictor profiles, and the distal link to suicidal-ideation outcomes are less well characterised in East Asian cohorts. Korean longitudinal trajectory studies of adolescent depression have largely used shorter follow-ups or specific subpopulations (Song et al., 2020), against a backdrop of considerable cross-sectional epidemiological work (Kwak and Ickovics, 2019); the seven-wave 2018 KCYPS panel, spanning the transition from early adolescence (age 13) to emerging adulthood (age 19), has not previously been exploited for trajectory enumeration focused on distal suicidality.

A second issue is criterion contamination. Most prior trajectory analyses estimated mixture models on composite depression scales that *include* a suicidal-ideation item, then reported associations between class membership and a later suicidal-ideation outcome built from the

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same item; the trajectory measure is therefore partly defined by the outcome being predicted, inflating distal effect sizes by a degree difficult to bound analytically. The KCYPS depression scale comprises ten items rated on a 1–4 Likert scale, of which E04 ("I have thoughts of dying") is the most face-valid suicidal-ideation indicator. A 9-item composite excluding E04 was used as the *primary* trajectory measure, with E04 retained separately as the distal outcome only; sensitivity analysis on the original 10-item composite gave Hungarian-aligned modal-class agreement $\kappa = 0.929$.

A developmental psychopathology and diathesis-stress perspective suggests that adolescent depressive symptoms should be understood by timing, persistence, and change across developmental transitions, not average level alone (Cicchetti and Rogosch, 2002). Sustained symptoms may confer risk through prolonged emotional burden, whereas later increases may become salient when school transition, stress exposure, or accumulating vulnerability activates previously less visible risk (Hankin et al., 2015; Monroe and Harkness, 2005). Escalating or chronic symptoms may increase SI through hopelessness, defeat, or entrapment (O'Connor and Kirtley, 2018), although these mechanisms are used here as an interpretive framework rather than tested mediators. This distinction motivates attention to both chronic and late-emerging pathways.

The KCYPS panel has two design features that shape the analyses reported below. Parenting at age 13 is captured with the Korean adaptation of the Skinner et al. (2005) Parents as Social Context Questionnaire, which partitions style into six dimensions (warmth, rejection, autonomy support, coercion, structure, and chaos), rather than collapsing parenting into a single bipolar dimension. The panel also carries annual ratings of self-esteem (Rosenberg, 1965), life satisfaction (Diener et al., 1985), grit (Duckworth and Quinn, 2009), peer attachment, teacher relationship, and smartphone dependence, and these were used here as a wide W1 covariate battery against which the resulting classes could be profiled.

This study therefore pursued four objectives, organised around a criterion-clean primary trajectory measure: (i) to enumerate latent depression trajectories from age 13 to 19 in the KCYPS 2018 m1 cohort, with the chosen K checked across two independently implemented optimisers (R lcmm and Mplus 7); (ii) to profile the resulting classes on a wide battery of W1 family, peer, school, and individual covariates; (iii) to estimate associations with W7 suicidal ideation, mean depression, and life satisfaction using the Bolck–Croon–Hagenaars (BCH) three-step correction; and (iv) to characterise time to first affirmative SI endorsement across all seven waves, with explicit attention to whether the proportional-hazards assumption holds and to how class-specific hazards evolve when it does not.

2. Methods

2.1. Sample and design

Data were drawn from the Korean Children and Youth Panel Survey 2018 (KCYPS 2018), middle-school first-grade (m1) cohort, conducted by the National Youth Policy Institute (National Youth Policy Institute, 2024). The original cohort comprised 2590 students entering middle-school grade 1 in 2018 and was sampled using two-stage stratified cluster sampling, with 17 provinces and three urban-size strata at the first stage and within-school student selection at the second stage. Surveys are conducted annually; the present analysis used data from Wave 1 (2018, age 13) through Wave 7 (2024, age 19). Sample retention was 94.1% at W2, 87.5% at W4, and 80.0% at W7.

The analytic sample (N = 2452, 94.7% of the original cohort) comprised individuals with at least three valid waves of the 9-item depression composite. Although LCGA estimation under full-information maximum likelihood (FIML) does not strictly require listwise data, a three-wave minimum was imposed so that each individual contributed to a substantive portion of the estimated trajectory. Sensitivity analysis with the W7 standardised longitudinal weight (WEIGHTB2w7) was performed to assess robustness to differential attrition. Throughout this paper, "sex" refers to the binary self-reported categorisation recorded in the KCYPS 2018 student roster; gender identity was not assessed in the panel, and the present analysis is restricted to that sex variable accordingly.

2.2. Measures

Depression — primary trajectory measure (9 items). The mean of YPSY4E01, E02, E03, E05, E06, E07, E08, E09, and E10 was computed for each wave (excluding E04 "I have thoughts of dying"). Items are rated on a four-point Likert scale (1 = *not at all* to 4 = *very much*). Internal consistency was high across all seven waves (Cronbach α between 0.88 and 0.91); the 9-item composite correlated r = 0.996 with the conventional 10-item composite at W1, indicating that exclusion of the suicidal-ideation item does not materially alter the construct for trajectory characterisation.

Depression — sensitivity (10 items). Mean of all ten YPSY4E items (α = 0.90–0.92), used in supplementary sensitivity analysis only.

Suicidal ideation (SI) — distal outcome. Item YPSY4E04 (Korean: '죽고 싶은 생각이 들었다'; English: 'I have thoughts of dying') was used as the distal outcome. Two response-category operationalisations were used: broad SI (≥ 2, 'sometimes or more') and affirmative SI endorsement (≥ 3, 'often or more'). Missing SI responses were retained as missing and were not coded as non-endorsement. For the time-to-event analysis, affirmative SI endorsement was used to identify first endorsement across W1-W7. Prior Korean adolescent research used the same KCYPS 2018 item as a single-item SI indicator (Donnelly et al., 2021), but did not provide criterion validation against a clinical assessment or validate the affirmative response category as a cutoff. These indicators are analytic response-category measures rather than diagnostic severity cutoffs.

Baseline (W1) covariates. Six dimensions of parenting were measured with the Korean adaptation of the Skinner et al. (2005) Parents as Social Context Questionnaire–Adolescent (warmth, rejection, autonomy support, coercion, structure, chaos; four items per dimension; α = 0.79–0.91). Other psychosocial variables included peer attachment (13 items adapted from the Inventory of Parent and Peer Attachment, Armsden and Greenberg, 1987; α = 0.85), teacher relationship (14 items developed by the National Youth Policy Institute for KCYPS; α = 0.91), Rosenberg self-esteem (10 items, α = 0.87), Diener Satisfaction With Life (5 items, α = 0.85), Lyubomirsky and Lepper (1999) subjective happiness (4 items, α = 0.80), Duckworth Short Grit (8 items, α = 0.71), and smartphone dependence (15 items adapted from the Smartphone Addiction Self-Diagnosis Scale of Kim et al., 2012; α = 0.88). Additional KCYPS-validated W1 psychosocial, behavioural, and academic-functioning scales used in the covariate profile and baseline-predictor analyses were aggression (6 items, α = 0.82), social withdrawal (5 items, α = 0.87), somatic symptoms (8 items, α = 0.85), attention problems (7 items, α = 0.85), academic engagement (9 items, α = 0.82), and academic helplessness (8 items, α = 0.87). Demographic indicators (sex, region, urban size) were taken from the W1 student roster.

2.3. Analytic strategy

2.3.1. Latent class growth analysis

Following Nagin (1999, 2005), finite mixtures of polynomial-in-time mean trajectories were specified with class-invariant residual variance and zero within-class growth-parameter variance, given by Eq. (1):

$$Y_{it} = \beta_{0k} + \beta_{1k}(t-4) + \beta_{2k}(t-4)^2 + \varepsilon_{it}, \varepsilon_{it} \sim N(0,\sigma^2) \quad (1)$$

where Wave 4 was used as the time origin. Estimation was via EM with FIML for missing waves, using eight random restarts. Solutions for K =

1–5 were compared on BIC, sample-size-adjusted BIC, AIC, entropy, smallest-class proportion, and a parametric bootstrap likelihood-ratio test (BLRT; Nylund et al., 2007; McLachlan, 1987) for K = 2 versus K = 1 with B = 30. Following Davison and Hinkley (1997), the BLRT *p*-value was reported under the add-one rule $p = (1 + \#\{\text{LRT_b} \geq \text{LRT_obs}\}) / (B + 1)$, so the smallest attainable p with B = 30 and zero exceedances is $1/31 \approx 0.032$; reaching $p < .001$ under the same rule would require $B \geq 1000$. BLRTs for larger K-vs-K-1 comparisons were not run because of computational cost, and enumeration beyond K = 2 relied on monotonic BIC reduction, entropy ≥0.66, and the 5%-minimum-class-size rule (Nagin, 2005).

2.3.2. Cross-validation in R and Mplus

K = 1–5 were re-fitted in R (lcmm 2.x; Proust-Lima et al., 2017; 30 gridsearch starts) and Mplus 7 (Muthén and Muthén, 1998–2017) with the same quadratic LCGA specification (growth-factor variances fixed to zero, residual variance class-invariant, time scores −3 to +3, STARTS = 1000 250). R lcmm uses a Newton–Marquardt optimiser; Mplus uses its own EM with random-start replication. Convergence across three independently implemented likelihoods provides triangulation against optimiser and software artefacts.

2.3.3. Bolck-Croon-Hagenaars three-step distal outcome correction

Posterior probabilities $P(C = s \mid Y_i)$ from the four-class LCGA were used to construct the classification-error matrix D, where D[s, t] denotes the conditional probability of modal assignment t given true class s. The BCH-corrected expectation of a distal outcome Y_{dist} for class s is then given by Eq. (2):

$$\hat{E}[Y_{dist} \mid C = s] = \Sigma_i \left(D^{-1}\right)[m_i, s]\, Y_{dist,i} / \Sigma_i \left(D^{-1}\right)[m_i, s] \quad (2)$$

where m_i is individual i's modal class. Standard errors and 95% confidence intervals were obtained by 10,000 nonparametric bootstrap resamples, with bootstrap proportions clipped to [0,1] to prevent boundary artefacts. The BCH approach (Bolck et al., 2004; Vermunt, 2010; Bakk and Vermunt, 2016) corrects for class-assignment uncertainty when entropy is moderate and is preferred over modal-class regression in this setting.

2.3.4. Time-to-event analysis

Time of first endorsement of affirmative SI across W1-W7 was the event; individuals with no endorsement were right-censored at the last observed wave. A Cox proportional-hazards model with class, wave, and class × wave interaction terms was used to characterise how hazards changed across calendar waves. Because class membership was estimated from the full Wave-1-7 depression course, these models are interpreted as descriptive calendar-time hazard characterisations rather than baseline predictive estimates; the SI item itself was excluded from the trajectory composite. The model was implemented with 'lifelines. CoxPHFitter' (Davidson-Pilon, 2019). The PH assumption was tested with Schoenfeld residuals; where it was violated, a time-varying class × wave specification was retained.

Because each individual contributes multiple wave-intervals to the long-form likelihood, the model-based covariance from 'variance_matrix_' is the Hessian inverse rather than a cluster-sandwich covariance, even when the model is fitted with cluster_col = 'ID', robust = True. Per-wave HR confidence intervals were therefore obtained via ID-level cluster bootstrap: B = 1000 resamples of unique individuals with replacement, with the time-varying Cox model refit on each resample. As with the BLRT, *p*-values were floored by the add-one rule at $1/(B + 1) \approx 0.001$.

2.3.5. Sensitivity analyses

Four sensitivity checks were performed in addition to the R lcmm and Mplus cross-validation. First, the LCGA was re-estimated using the 10-item depression composite as the trajectory measure; modal-class agreement with the primary solution was assessed after Hungarian label alignment. Second, the LCGA was re-fitted with the W7 longitudinal weight. Third, modal-class logistic models were compared with BCH estimates to gauge attenuation from classification error. Fourth, covariate-adjusted modal-class logistic models controlled for sex, household income, paternal education, and maternal education to examine whether the C4 versus C2 contrast was robust to demographic and socioeconomic adjustment.

2.3.6. W1 covariate comparisons and R3STEP baseline predictor model

Class profiles on Wave 1 covariates (Supplementary Table S5) were compared using one-way ANOVA on modal class assignments. This treats the modal class as if it were observed, and therefore does not propagate LCGA classification uncertainty into standard errors (Vermunt, 2010). To address this limitation, baseline covariates were also examined using the Mplus R3STEP multinomial approach, which accounts for classification uncertainty when estimating independent baseline correlates of class membership (Asparouhov and Muthén, 2014). Continuous predictors were standardised, complete cases on auxiliary variables were used (N = 2237), and false-discovery-rate correction was applied.

2.4. Software and reproducibility

All analyses were conducted in Python 3.11 (numpy, pandas, scipy, statsmodels, scikit-learn, lifelines), R 4.5.2 (lcmm 2.x, haven, dplyr, tidyr), and Mplus 7. Random seeds were fixed ('seed = 42') for EM restarts and bootstrap resampling. Total run-time on a single CPU was approximately 5 min for the LCGA, 6 min for the BLRT, 30 s for the cluster bootstrap, 30 min for the R lcmm gridsearch, and a few minutes per K for the K = 1–5 Mplus runs (STARTS = 1000 250).

2.5. Ethics

This study analysed de-identified public-use microdata released by the National Youth Policy Institute (NYPI), Republic of Korea, and is therefore exempt from institutional review board approval as a non-human-subjects research activity under standard secondary-use guidelines. The original Korean Children and Youth Panel Survey 2018 was approved by the NYPI institutional review process, and written informed consent was obtained from all participants and their legal guardians at the time of original data collection.

3. Results

3.1. Sample characteristics

Sociodemographic and W1 psychosocial characteristics of the analytic sample are summarised in Supplementary Table S3. The 2452 retained adolescents were 46.2% female, with class-specific female proportions ranging from 40.3% in Late-Increasing and 40.5% in Stable-Low to 53.3% in Decreasing and 60.4% in Persistent-High ($\chi^2(3) = 54.8$, $p < .001$). At W1, depression mean was 1.83 (SD 0.65) on the 9-item primary scale, and the four classes differed substantially on this measure ($F = 687.0$, $p < .001$, $\eta^2 = 0.457$). Class percentages in Supplementary Table S3 are modal-class proportions (modal n / N: 25.1%, 10.0%, 29.6%, 35.3%) and differ slightly from the EM mixing proportions $\hat{\pi}$ in Supplementary Table S4 (25.4%, 10.5%, 29.3%, 34.8%); throughout, "class size" refers to modal n while distal-outcome estimation uses the BCH classification-error matrix.

3.2. Number of classes

Table 1 reports fit indices for the 9-item primary LCGA across K = 1 to 5. BIC decreased monotonically (28,317.6 to 25,126.6), as is common in finite-mixture models, so BIC alone was not treated as sufficient

Table 1
Latent class growth analysis fit indices and class-size diagnostics for K = 1 through K = 5 on the 9-item depression composite (analytic $N = 2452$).

K	log L	BIC	aBIC	AIC	Entropy	Smallest class %
1	−14,143.19	28,317.60	28,304.89	28,294.38	1.000	100.0%
2	−13,055.84	26,174.13	26,148.71	26,127.69	0.675	41.4%
3	−12,746.20	25,586.06	25,547.93	25,516.40	0.719	12.3%
4	−12,551.60	25,228.08	25,177.24	25,135.20	0.661	10.0%
5	−12,485.23	25,126.55	25,063.01	25,010.46	0.667	3.5%

Note. K = number of latent classes. BIC = Bayesian information criterion; aBIC = sample-size-adjusted BIC; AIC = Akaike information criterion. The BLRT was estimated only for K = 2 versus K = 1 ($p = .032$) and is not used as the basis for higher-order class enumeration. Adjacent-class TECH11 results are reported in Supplementary Table S1. The four-class solution was retained based on information criteria, class size, entropy, posterior classification quality, adjacent-class tests, and interpretability.

evidence for the five-class solution. The four-class solution was selected by combining information criteria, entropy, smallest-class size, posterior classification quality, adjacent-class likelihood-ratio tests, and substantive interpretability.

Independent re-fitting of K = 1–5 in R (lcmm 2.x, 30 gridsearch starts) and Mplus 7 (quadratic LCGA, STARTS = 1000 250) returned essentially identical fit indices to the Python EM solution: K = 4 BIC = 25,228.08 in both Python and Mplus, 25,228.10 in R. TECH11 tests supported improvement from three to four classes (VLMR-LRT $p < .001$; adjusted LMR p < .001), whereas the five-class solution did not significantly improve fit over the four-class solution (VLMR $p = .101$; adjusted LMR $p = .107$) and produced a small additional class below the 5% rule. K = 3 did not separate the rising and decreasing patterns, whereas K = 5 added limited substantive interpretability. The K = 4 solution was therefore retained. Supplementary Table S1 and Supplementary Figs. S1 and S3 provide the adjacent-class tests and fit-index diagnostics.

3.3. Class profiles

The four model-implied class trajectories are depicted in Fig. 1 and summarised numerically in Supplementary Table S4. *Stable-Low/Resilient* ($\hat{\pi} = 25.4\%$; modal $n = 615$) showed an intercept of 1.33 and a near-zero linear slope, with depression mean 1.32–1.46 across waves. *Persistent-High* ($\hat{\pi} = 10.5\%$; $n = 245$) had an intercept of 2.61 with a slight inverted-U shape (peak 2.62 at W3); this is the chronically symptomatic class. *Decreasing* ($\hat{\pi} = 29.3\%$; $n = 726$) had intercept 1.93 and slope − 0.10, declining from 2.28 at W1 to 1.68 at W7; a recovering trajectory whose risk profile resembles *Persistent-High* in early adolescence but converges toward *Stable-Low* by emerging adulthood. *Late-Increasing* ($\hat{\pi} = 34.8\%$; $n = 866$) is the largest class and shows the inverse pattern: intercept 1.86, slope + 0.07, with depression rising from 1.49 at W1 (essentially indistinguishable from *Stable-Low*) to 1.93 at W7.

Persistent-High and *Decreasing* show essentially overlapping mean depression at W1 (2.39 vs 2.28) and are separated primarily by their *trajectory shape* (the chronic/inverted-U pattern of C2 versus the linear decline of C3) rather than by W1 mean level alone. The same logic

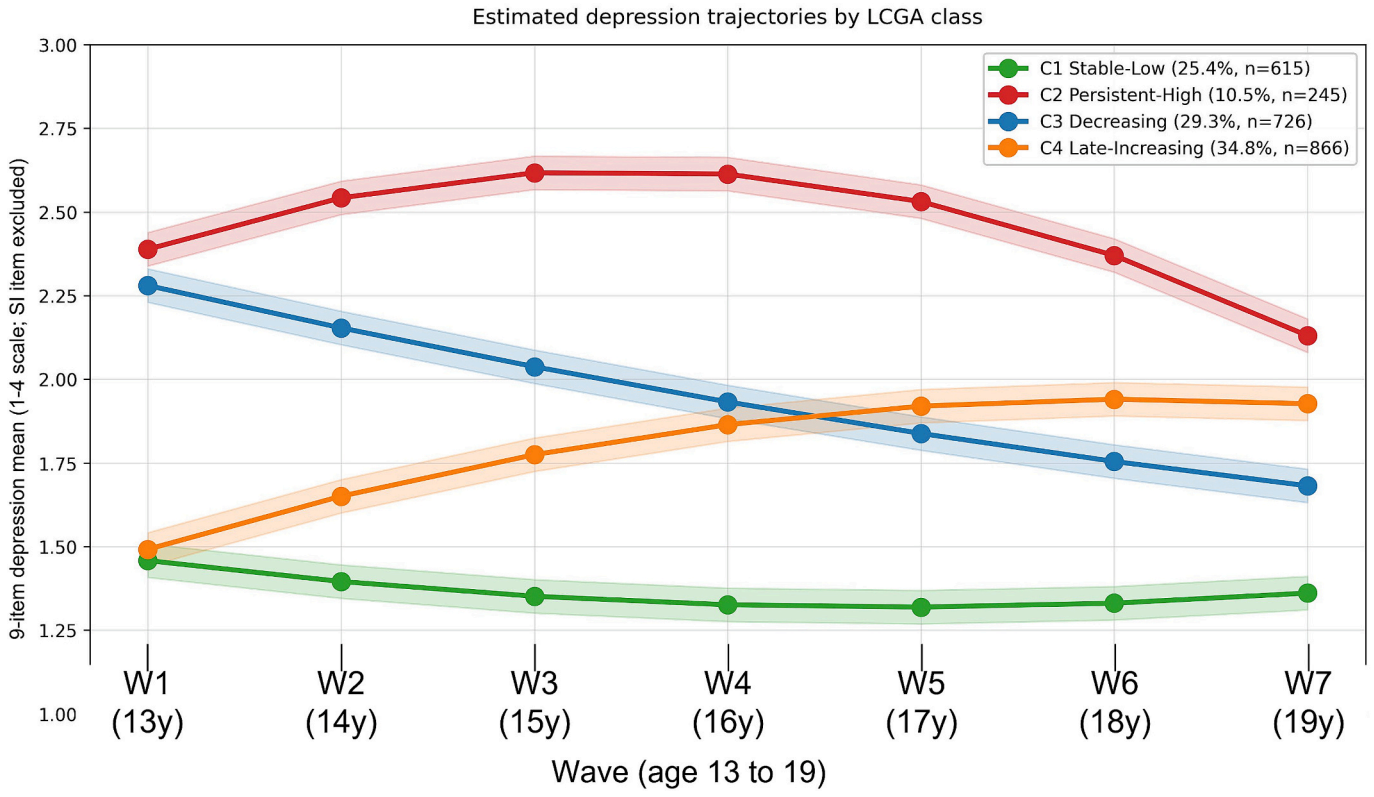


Fig. 1. Estimated mean depression trajectories across Waves 1–7 (age 13 to 19) for the four-class LCGA solution on the 9-item depression composite (criterion-clean; suicidal-ideation item E04 excluded). Class labels: C1 *Stable-Low/Resilient* (modal $n = 615$; $\hat{\pi} = 25.4\%$), C2 *Persistent-High* ($n = 245$; $\hat{\pi} = 10.5\%$), C3 *Decreasing* ($n = 726$; $\hat{\pi} = 29.3\%$), C4 *Late-Increasing* ($n = 866$; $\hat{\pi} = 34.8\%$). Y-axis: depression mean on 1–4 Likert scale.

Table 2
Bolck-Croon-Hagenaars three-step corrected distal outcomes at Wave 7 (age 19) by trajectory class. Bootstrap 95% percentile confidence intervals from B = 10,000 resamples.

Outcome (W7)	Class or contrast	BCH-corrected estimate [95% CI]	OR or contrast [95% CI]
Broad SI (≥2)	C1 Stable-Low	9.4% [5.5, 13.4]	1.00 (ref)
	C2 Persistent-High	56.7% [48.5, 64.7]	12.63 [7.55, 24.43]
	C3 Decreasing	36.8% [31.3, 42.3]	5.61 [3.52, 10.46]
	C4 Late-Increasing	66.2% [61.3, 71.3]	18.86 [11.51, 36.05]
	C4 vs C2	–	1.49 [1.00, 2.25], p = .052
Affirmative SI (≥3)	C1 Stable-Low	1.3% [0.0, 3.1]	1.00 (ref)
	C2 Persistent-High	16.1% [10.6, 22.1]	14.97 [5.36, 165.08]
	C3 Decreasing	2.9% [0.5, 5.4]	2.33 [0.38, 24.84]
	C4 Late-Increasing	13.4% [10.4, 16.6]	12.04 [4.32, 136.77]
	C4 vs C2	–	0.80 [0.50, 1.38], p = .426
Mean depression (1–4)	C1 Stable-Low	1.17 [1.13, 1.21]	–
	C2 Persistent-High	2.19 [2.09, 2.28]	–
	C3 Decreasing	1.53 [1.48, 1.59]	–
	C4 Late-Increasing	2.04 [1.99, 2.09]	–
Life satisfaction (1–4)	C1 Stable-Low	2.73 [2.68, 2.79]	–
	C2 Persistent-High	2.32 [2.24, 2.40]	–
	C3 Decreasing	2.53 [2.48, 2.58]	–
	C4 Late-Increasing	2.49 [2.45, 2.54]	–

Note. BCH = Bolck-Croon-Hagenaars three-step distal outcome correction (Bolck et al., 2004; Vermunt, 2010). Classification-error matrix D constructed from posterior probabilities of the four-class LCGA solution. Missing SI responses were retained as missing and were not coded as non-endorsement. Broad SI is E04 ≥ 2; affirmative SI endorsement is E04 ≥ 3. Direct C4 versus C2 contrasts are included to evaluate whether Late-Increasing differs from Persistent-High. Affirmative-SI odds ratios versus Stable-Low have wide confidence intervals because affirmative SI was rare in the reference class and should be interpreted cautiously.

applies in mirror image to *Late-Increasing*, which overlaps with *Stable-Low* at W1 and is separated only by the rising trajectory across subsequent waves; two of the four developmental subgroups are thus not readily identifiable on Wave 1 information alone.

After Hungarian alignment to the manuscript labels, the R lcmm and Mplus solutions matched the Python EM modal assignments to within ±1 individual per class (class-count alignment across platforms is reported in Supplementary Table S2).

3.4. Baseline (W1) predictors of class membership

All eighteen W1 psychosocial constructs differed significantly across the four trajectory classes (Supplementary Table S5; all $F > 36$, $p < .001$, well below a Bonferroni-adjusted alpha = 0.05/18 = 0.003). Largest effects involved depression itself, self-esteem, somatic symptoms, subjective happiness, life satisfaction, and social withdrawal. In the R3STEP multinomial model (Supplementary Table S10), baseline covariates more clearly distinguished the Persistent-High and Decreasing classes than the Late-Increasing class. After false-discovery-rate correction, the Late-Increasing class was not robustly distinguished from Stable-Low by Wave-1 covariates, particularly after accounting for baseline depressive symptoms. This pattern supports the interpretation that late-emerging risk may be difficult to identify using one-time baseline screening alone. Supplementary Fig. S2 visualises the standardised covariate profile.

3.5. Distal outcomes at Wave 7

BCH-corrected distal outcomes are reported in Table 2 (modal-class comparison in Section 3.7); Fig. 2 shows wave-by-wave endorsement (observed rates and missingness in Supplementary Table S8), and Fig. 3 forest-plots Wave-7 ORs versus Stable-Low and the direct Late-Increasing versus Persistent-High contrast. Broad SI at Wave 7 was lowest in Stable-Low (9.4%, 95% CI 5.5%–13.4%) and substantially elevated in Persistent-High (56.7%, 95% CI 48.5%–64.7%) and Late-Increasing (66.2%, 95% CI 61.3%–71.3%). Relative to Stable-Low, odds of broad SI were higher in Persistent-High (OR = 12.63, 95% CI 7.55–24.43), Decreasing (OR = 5.61, 95% CI 3.52–10.46), and Late-Increasing (OR = 18.86, 95% CI 11.51–36.05). The direct Late-Increasing versus Persistent-High contrast was borderline for broad SI (OR = 1.49, 95% CI 1.00–2.25, $p = .052$) and not conventionally significant for affirmative SI (OR = 0.80, 95% CI 0.50–1.38, $p = .426$). These results do not support ranking Late-Increasing above Persistent-High; they indicate two high-risk pathways with different timing.

3.6. Time to first affirmative suicidal ideation

Across the seven-year observation window, 962 of 2452 individuals (39.2%) endorsed affirmative SI at some wave; class-specific cumulative event rates were 13.5% (Stable-Low), 38.8% (Late-Increasing), 49.3% (Decreasing), and 65.7% (Persistent-High). Because class membership was estimated from the full W1-W7 depression course, these values characterise calendar-time hazard patterns rather than baseline predictive estimates.

Schoenfeld tests rejected PH for Decreasing and Late-Increasing (both $p < .001$), motivating a time-varying class × wave Cox model (Table 3, Supplementary Table S6, Supplementary Fig. S4).

Persistent-High hazards remained uniformly elevated (HR W1 = 12.35, 95% CI 8.10–19.59; HR W4 = 9.02, 95% CI 7.00–12.03; HR W7 = 6.59, 95% CI 3.67–12.18), with a non-significant class × wave interaction (beta = −0.105, 95% CI -0.253 to 0.046, $p = .196$). Decreasing hazards fell (HR W1 = 8.96, 95% CI 6.04–14.10; HR W7 = 1.37, 95% CI 0.73–2.57; beta = −0.313, 95% CI -0.469 to −0.166, $p < .001$), while Late-Increasing hazards rose from non-significant at W1 to elevated by W7 (HR W1 = 1.43, 95% CI 0.95–2.32; HR W4 = 3.24, 95% CI 2.60–4.20; HR W7 = 7.34, 95% CI 4.60–12.96; beta = 0.272, 95% CI 0.138–0.415, $p < .001$).

3.7. Sensitivity analyses

Re-estimating the LCGA on the 10-item composite (E04 included) recovered the same four-class structure (modal-class κ = 0.929 after Hungarian alignment); the Late-Increasing broad-SI BCH OR remained elevated, demonstrating criterion-contamination robustness in the original direction. Re-fitting with the W7 standardised longitudinal weight (WEIGHTB2w7) reproduced the four-class shapes (κ = 0.939).

Modal-class logistic odds ratios were smaller than BCH-corrected estimates, consistent with attenuation from classification error. In covariate-adjusted modal-class sensitivity models controlling for sex, household income, paternal education, and maternal education, the direct Late-Increasing versus Persistent-High contrast was not statistically significant for broad SI (OR = 1.16, 95% CI 0.84–1.59, $p = .369$) or

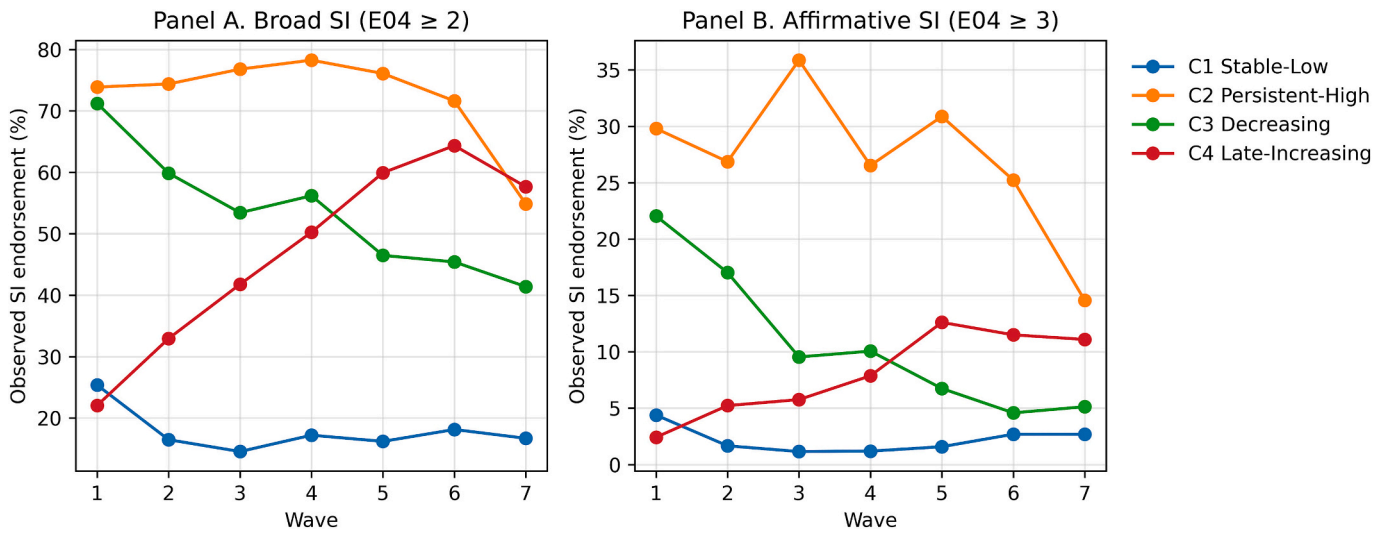


Fig. 2. Suicidal-ideation endorsement by depression trajectory class across all seven waves of the 9-item primary analysis. Panel A shows broad SI (E04 ≥ 2), and Panel B shows affirmative SI endorsement (E04 ≥ 3). Missing SI responses were retained as missing rather than coded as non-endorsement. The W4-W5 crossover between Late-Increasing and Decreasing is most apparent in Panels A-B.

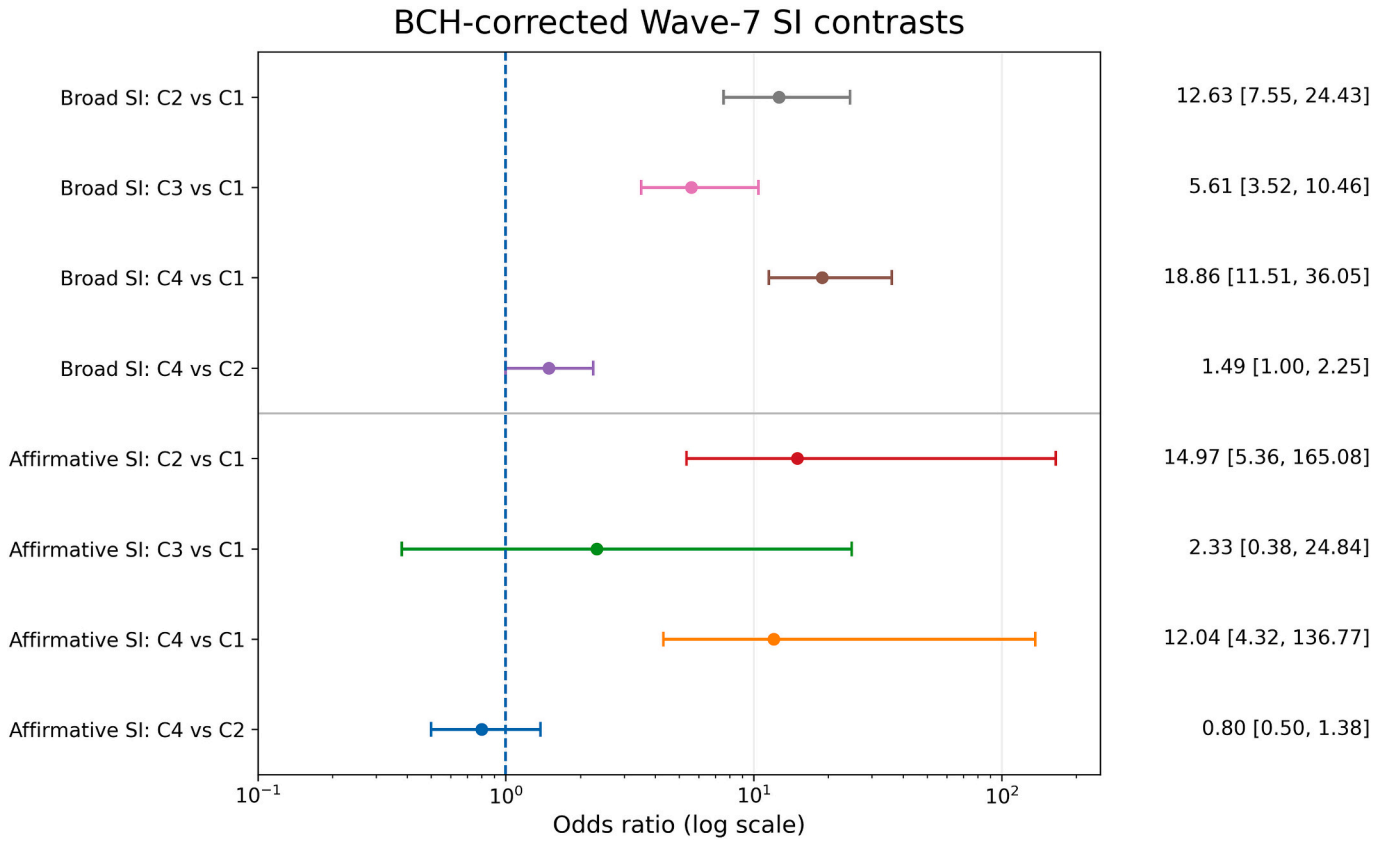


Fig. 3. Forest plot of Wave-7 Bolck-Croon-Hagenaars-corrected odds ratios for broad and affirmative SI (B = 10,000 percentile bootstrap; effects plotted on the log scale). The figure includes contrasts against the Stable-Low reference class and the direct Late-Increasing versus Persistent-High contrast; the latter is interpreted cautiously rather than as a definitive hierarchy of risk. Wide CIs for affirmative-SI ORs reflect small reference-cell counts. Numerical OR estimates are displayed alongside each point estimate; corresponding first-affirmative-SI hazard estimates are reported in Table 3, Supplementary Table S6, and Supplementary Fig. S4.

affirmative SI (OR = 0.79, 95% CI 0.50–1.26, $p = .323$). This sensitivity check reinforces the interpretation of Persistent-High and Late-Increasing as two high-risk trajectories rather than a definitive hierarchy of risk. Full adjusted models are reported in Supplementary Table S9.

4. Discussion

4.1. Principal findings

A four-class developmental typology of Korean adolescent depression (age 13–19) distinguishes resilient (25%), recovering (29%), late-

Table 3
Per-wave hazard ratios for first endorsement of affirmative suicidal ideation (W1-W7) under the time-varying class × wave Cox model, with ID-level cluster-bootstrap 95% percentile confidence intervals (B = 1000).

Class (vs C1 Stable-Low)	HR W1 [95% CI]	HR W4 [95% CI]	HR W7 [95% CI]	Class × wave beta [95% CI]	p
C2 Persistent-High	12.35 [8.10, 19.59]	9.02 [7.00, 12.03]	6.59 [3.67, 12.18]	−0.105 [−0.253, 0.046]	.196
C3 Decreasing	8.96 [6.04, 14.10]	3.51 [2.69, 4.61]	1.37 [0.73, 2.57]	−0.313 [−0.469, −0.166]	<.001
C4 Late-Increasing	1.43 [0.95, 2.32]	3.24 [2.60, 4.20]	7.34 [4.60, 12.96]	0.272 [0.138, 0.415]	<.001

Note. Schoenfeld residual tests rejected the proportional-hazards assumption for Decreasing and Late-Increasing. HR = hazard ratio. CI = confidence interval. P-values use the add-one bootstrap floor at 1/(B + 1) = 0.001. The class × wave terms are shown in the table body because they test whether class differences change across waves. Full seven-wave estimates and discrete-time cloglog sensitivity are reported in Supplementary Table S6.

rising (35%), and chronically high (10%) trajectories. Corrected BCH estimates show elevated Wave-7 broad SI in both Persistent-High and Late-Increasing relative to Stable-Low, while the direct Late-Increasing versus Persistent-High contrast was borderline rather than conventionally significant. The central interpretation is therefore refined rather than overturned: Persistent-High and Late-Increasing represent two high-risk pathways, one characterised by sustained depressive burden and the other by late-emerging escalation. These results rest on a criterion-clean depression composite that excludes the suicidal-ideation item and on BCH three-step correction of distal estimates.

4.2. The "invisible at-risk" finding

At Wave 1, Late-Increasing was close to Stable-Low on baseline depression and was not robustly distinguished from Stable-Low on the measured W1 covariates. By Wave 7 it reaches the highest point estimate for broad SI across the four classes (66.2%), but the direct comparison with Persistent-High is borderline and should not be read as evidence that Late-Increasing has clearly higher SI risk than Persistent-High. Its hazard pattern nevertheless rises across adolescence (class × wave beta = 0.272, $p < .001$), whereas Persistent-High shows sustained elevation throughout the observation window. This pattern is consistent with an emerging-vulnerability pathway in which risk becomes more visible during later adolescence, as developmental and school-transition demands accumulate beyond available coping resources, rather than being fully apparent at school entry.

The R3STEP baseline predictor model provides a more conservative basis for this screening interpretation than the original modal-class ANOVA. Persistent-High and Decreasing were more clearly distinguished from Stable-Low by Wave-1 covariates, including female sex and internalising-related indicators. Late-Increasing, however, was not robustly distinguished from Stable-Low after false-discovery-rate correction, particularly after accounting for baseline depressive symptoms. The findings support the value of repeated assessment across the high-school transition as a complement to single-time-point screening, and stand in contrast to Persistent-High, which is more visible already at age 13 through elevated depressive burden.

4.3. Comparison with international literature

Class proportions in Western adolescent cohorts (Stoolmiller et al., 2005; Costello et al., 2008; Yaroslavsky et al., 2013) typically show a larger Stable-Low class (≈ 50%) and a smaller Increasing class (≈ 10–15%); the present *Late-Increasing* proportion of ≈ 35% is unusually large by these comparators. Possible drivers include the academic-track stratification of the Korean high-school system, examination pressure surrounding the College Scholastic Ability Test, rapid digital-media integration, and differences in school-based mental-health support and help-seeking pathways; differences in measurement, follow-up duration, and cohort recruitment may also contribute.

4.4. Parenting and other predictors

All six Skinner parenting dimensions differentiated classes at age 13 (η^2 = 0.04–0.09), but were weaker differentiators than internalising indicators (depression itself $\eta^2 = 0.457$; self-esteem $\eta^2 = 0.247$; somatic symptoms $\eta^2 = 0.237$). Using the observed modal-class W1 means in Supplementary Table S5, the Late-Increasing-Stable-Low gap in baseline depression was only 0.04 points (Late-Increasing 1.46 vs Stable-Low 1.50; Cohen d = −0.10), far too small to serve as a clinical screening cutoff ($|d| < 0.15$ across 18 covariates). The R3STEP results similarly indicate that Wave-1 covariates did not robustly distinguish Late-Increasing from Stable-Low after correction, whereas Persistent-High and Decreasing showed clearer baseline differences. Sex composition differed sharply: Persistent-High (60.4% female) and Decreasing (53.3% female) align with known adolescent female-onset patterns, whereas Late-Increasing (40.3% female) was not female-dominant, suggesting that this trajectory is not simply an extension of the conventional female-excess pattern in adolescent depression.

4.5. Methodological strengths

Strengths include the criterion-clean 9-item specification, 10-item sensitivity replication ($\kappa = 0.929$), BCH correction with ID-level bootstrap, and cross-platform replication across Python, R lcmm, and Mplus 7 ($\kappa = 0.999$).

4.6. Limitations

Reported associations are *associative, not causal*: LCGA describes developmental heterogeneity in a single observational cohort, and the BCH and Cox estimates quantify class–outcome associations adjusted for classification uncertainty.

Class enumeration and entropy. BIC declined monotonically through K = 5, so the four-class solution was retained using adjacent-class tests, parsimony, class size, posterior classification quality, and interpretability. VLMR and adjusted LMR tests supported K = 4 over K = 3 (both $p < .001$) but not K = 5 over K = 4 ($p = .101$ and .107; Supplementary Table S1). Entropy was moderate (0.66); BCH and R3STEP reduce, but do not remove, classification-uncertainty concerns, so classes should be interpreted as empirical trajectory patterns rather than sharply bounded clinical subtypes.

Outcome and event-time measurement. Depressive symptoms and SI were self-reported, SI was measured with a single item rather than a clinical interview, and clinical diagnoses were unavailable. Broad and affirmative SI are response-category operationalisations rather than validated diagnostic cutoffs; common-method bias may have inflated self-report associations. Exact event time within annual intervals is unknown, but a discrete-time cloglog sensitivity model returned interaction coefficients close to the Cox specification. Because classes were estimated from the full Wave-1-7 course, repeated-SI analyses are descriptive calendar-time characterisations rather than baseline prospective prediction.

Attrition and model assumptions. W7 retention was 80%, with analysis restricted to the 94.7% of the original cohort having ≥3 valid waves (138 of 2590 W1 respondents excluded). Across 21 W1 covariates, most showed small differences between excluded and analytic

samples; only aggression and academic achievement reached approximately $|d| = 0.20$, and W1 depression itself was $|d| = 0.11$ (Supplementary Table S7). Re-fitting under the W7 longitudinal weight reproduced the four-class structure ($\kappa = 0.939$). LCGA imposes class-invariant residual variance and zero within-class growth-parameter variance; growth mixture models with class-varying random effects can capture finer heterogeneity at greater computational cost (Frankfurt et al., 2016).

Class label and rare-outcome estimates. Late-Increasing refers to trajectory shape: in the model-implied trajectory, its W7 depression mean (1.93) remained above Stable-Low (1.36) but below Persistent-High (2.13). BCH bootstrap CIs on affirmative SI are wide because affirmative SI was rare in the Stable-Low reference class.

Period effects. The COVID-19 pandemic overlapped Waves 3 (2020) through 5 (2022), and unmeasured period effects on the *Late-Increasing* pattern cannot be ruled out. However, inspection of Fig. 1 indicates that the *Late-Increasing* slope was already positive before W3 and plateaued after W5, suggesting that the rising pattern preceded the pandemic onset rather than tracking it. The *Persistent-High* and *Decreasing* classes traversed the same calendar interval with sustained elevation and pronounced decline respectively, suggesting that a uniform period-driven mean shift is an incomplete account; period and class-specific developmental processes likely co-occur.

Sensitivity analyses not undertaken. Sex-stratified LCGA was not estimated separately given the small size of certain class strata (notably Persistent-High $n = 245$). Covariate-adjusted modal-class sensitivity models were added, but fully adjusted BCH distal models were not estimated because they lie outside the standard BCH implementation used here. The trajectory class spans all seven waves and so partly overlaps the repeated SI outcome in calendar time; a strict-prospective sensitivity analysis (Wave 1–6 trajectories with Wave 7 ideation as a future outcome) was not conducted. These issues are flagged as future work.

4.7. Practical implications

To the extent that school-based depression screening relies on a single assessment, the results suggest that it may miss adolescents whose risk emerges later. Persistent-High may require early sustained support, whereas Late-Increasing may require monitoring that continues into later adolescence as symptoms escalate. For Late-Increasing, intervention implications extend beyond screening to reducing accumulated stress, strengthening coping resources, and connecting adolescents to family-, school-, or community-based services. These suggestions are based on one observational cohort and require replication before policy adoption; trajectory class is a post hoc summary, not a Wave-1 predictor.

5. Conclusions

A criterion-clean four-class developmental typology of Korean adolescent depression, estimated across seven annual waves from age 13 to age 19, indicates that Persistent-High and Late-Increasing represent two high-risk pathways with different timing. Persistent-High reflects sustained depressive burden and elevated SI risk across adolescence, whereas Late-Increasing reflects a late-emerging pathway whose risk becomes more visible over time. Time-varying hazard modelling shows this timing distinction without supporting a causal or Wave-1 predictive interpretation. Together with the seven-wave longitudinal design, BCH correction, cross-platform LCGA replication, R3STEP baseline-predictor analysis, and covariate-adjusted sensitivity checks, these findings suggest that depression trajectories contribute information beyond single-wave symptom levels for identifying adolescents who later show elevated SI. Replication and prospective evaluation of screening and intervention implications may be warranted.

CRediT authorship contribution statement

Joonsung Kwon: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used Claude to support language editing, structural feedback, and manuscript-formatting checks. After using these tools, the author reviewed and edited all output and takes full responsibility for the content of the published article. AI tools were not used to generate, interpret, or analyse the underlying data, nor to draw scientific conclusions. Generative AI tools were not used to create or alter any figures, images, or artwork.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2026.122306.

Data availability

KCYPS 2018 public-use microdata are available from the National Youth Policy Institute upon registration at https://www.nypi.re.kr/archive/.

References

Armsden, G.C., Greenberg, M.T., 1987. The Inventory of Parent and Peer Attachment: individual differences and their relationship to psychological well-being in adolescence. J. Youth Adolesc. 16, 427–454. https://doi.org/10.1007/BF02202939.

Asparouhov, T., Muthén, B., 2014. Auxiliary variables in mixture modeling: three-step approaches using Mplus. Struct. Equ. Model. 21, 329–341. https://doi.org/10.1080/10705511.2014.915181.

Bakk, Z., Vermunt, J.K., 2016. Robustness of stepwise latent class modeling with continuous distal outcomes. Struct. Equ. Model. 23, 20–31. https://doi.org/10.1080/10705511.2014.955104.

Bolck, A., Croon, M., Hagenaars, J., 2004. Estimating latent structure models with categorical variables: one-step versus three-step estimators. Polit. Anal. 12, 3–27. https://doi.org/10.1093/pan/mph001.

Cicchetti, D., Rogosch, F.A., 2002. A developmental psychopathology perspective on adolescence. J. Consult. Clin. Psychol. 70, 6–20. https://doi.org/10.1037/0022-006X.70.1.6.

Costello, D.M., Swendsen, J., Rose, J.S., Dierker, L.C., 2008. Risk and protective factors associated with trajectories of depressed mood from adolescence to early adulthood. J. Consult. Clin. Psychol. 76, 173–183. https://doi.org/10.1037/0022-006X.76.2.173.

Davidson-Pilon, C., 2019. lifelines: survival analysis in Python. J. Open Source Softw. 4, 1317. https://doi.org/10.21105/joss.01317.

Davison, A.C., Hinkley, D.V., 1997. Bootstrap Methods and Their Application. Cambridge University Press, Cambridge.
Dekker, M.C., Ferdinand, R.F., van Lang, N.D.J., Bongers, I.L., van der Ende, J., Verhulst, F.C., 2007. Developmental trajectories of depressive symptoms from early childhood to late adolescence. J. Child Psychol. Psychiatry 48, 657–666. https://doi.org/10.1111/j.1469-7610.2007.01742.x.
Diener, E., Emmons, R.A., Larsen, R.J., Griffin, S., 1985. The satisfaction with life scale. J. Pers. Assess. 49, 71–75. https://doi.org/10.1207/s15327752jpa4901_13.
Donnelly, H.K., Richardson, D., Solberg, S.V., 2021. Understanding somatic symptoms associated with South Korean adolescent suicidal ideation, depression, and social anxiety. Behav. Sci. 11, 151. https://doi.org/10.3390/bs11110151.
Duckworth, A.L., Quinn, P.D., 2009. Development and validation of the Short Grit Scale (Grit-S). J. Pers. Assess. 91, 166–174. https://doi.org/10.1080/00223890802634290.
Frankfurt, S., Frazier, P., Syed, M., Jung, K.R., 2016. Using group-based trajectory and growth mixture modeling to identify classes of change trajectories. Couns. Psychol. 44, 622–660. https://doi.org/10.1177/0011000016658097.
Hankin, B.L., Young, J.F., Abela, J.R.Z., Smolen, A., Jenness, J.L., Gulley, L.D., Technow, J.R., Gottlieb, A.B., Cohen, J.R., Oppenheimer, C.W., 2015. Depression from childhood into late adolescence: influence of gender, development, genetic susceptibility, and peer stress. J. Abnorm. Psychol. 124, 803–816. https://doi.org/10.1037/abn0000089.
Kim, D., Chung, Y., Lee, J., Kim, M., Lee, Y., Kang, E., Keum, C., Nam, J., 2012. Development of a brief smartphone addiction self-diagnosis scale for adults. Korea J. Couns. 13, 629–644. https://doi.org/10.15703/kjc.13.2.201204.629 (in Korean).
Kim, H., Lee, Y.J., Jung, S.J., 2023. Trends in depressive symptoms and suicidality of South Korean adolescents: comparison of expected and observed prevalence during the COVID-19 pandemic. J. Adolesc. Health 73, 79–87. https://doi.org/10.1016/j.jadohealth.2023.02.014.
Kwak, C.W., Ickovics, J.R., 2019. Adolescent suicide in South Korea: risk factors and proposed multi-dimensional solution. Asian J. Psychiatr. 43, 150–153. https://doi.org/10.1016/j.ajp.2019.05.027.
Lyubomirsky, S., Lepper, H.S., 1999. A measure of subjective happiness: preliminary reliability and construct validation. Soc. Indic. Res. 46, 137–155. https://doi.org/10.1023/A:1006824100041.
McLachlan, G.J., 1987. On bootstrapping the likelihood ratio test statistic for the number of components in a normal mixture. Appl. Stat. 36, 318–324. https://doi.org/10.2307/2347790.
Monroe, S.M., Harkness, K.L., 2005. Life stress, the kindling hypothesis, and the recurrence of depression: considerations from a life stress perspective. Psychol. Rev. 112, 417–445. https://doi.org/10.1037/0033-295X.112.2.417.
Muthén, L.K., Muthén, B.O., 1998–2017. Mplus User's Guide, Eighth ed. Muthén & Muthén, Los Angeles, CA.
Nagin, D.S., 1999. Analyzing developmental trajectories: a semiparametric, group-based approach. Psychol. Methods 4, 139–157. https://doi.org/10.1037/1082-989X.4.2.139.
Nagin, D.S., 2005. Group-based Modeling of Development. Harvard University Press, Cambridge, MA.
National Youth Policy Institute, 2024. Korean Children and Youth Panel Survey 2018 (KCYPS 2018), Wave 1–7 Public-use Microdata. NYPI, Sejong, Republic of Korea. https://www.nypi.re.kr/archive/.
Nylund, K.L., Asparouhov, T., Muthén, B.O., 2007. Deciding on the number of classes in latent class analysis and growth mixture modeling: a Monte Carlo simulation study. Struct. Equ. Model. 14, 535–569. https://doi.org/10.1080/10705510701575396.
O'Connor, R.C., Kirtley, O.J., 2018. The integrated motivational-volitional model of suicidal behaviour. Philos. Trans. R. Soc. Lond. Ser. B Biol. Sci. 373, 20170268. https://doi.org/10.1098/rstb.2017.0268.
Park, E., Lee, H.Y., 2024. The residential, area-specific prevalence and associated factors of suicidal ideation among South Korean adolescents. Prev. Med. Rep. 41, 102698. https://doi.org/10.1016/j.pmedr.2024.102698.
Proust-Lima, C., Philipps, V., Liquet, B., 2017. Estimation of extended mixed models using latent classes and latent processes: the R package lcmm. J. Stat. Softw. 78, 1–56. https://doi.org/10.18637/jss.v078.i02.
Rosenberg, M., 1965. Society and the Adolescent Self-image. Princeton University Press, Princeton.
Skinner, E., Johnson, S., Snyder, T., 2005. Six dimensions of parenting: a motivational model. Parenting 5, 175–235. https://doi.org/10.1207/s15327922par0502_3.
Song, M.K., Yoon, J.Y., Kim, E., 2020. Trajectories of depressive symptoms among multicultural adolescents in Korea: longitudinal analysis using latent class growth model. Int. J. Environ. Res. Public Health 17, 8217. https://doi.org/10.3390/ijerph17218217.
Statistics Korea, 2024. Annual Report on the Causes of Death Statistics 2023. Statistics Korea, Daejeon. https://kostat.go.kr/. (Accessed 1 March 2026).
Stoolmiller, M., Kim, H.K., Capaldi, D.M., 2005. The course of depressive symptoms in men from early adolescence to young adulthood: identifying latent trajectories and early predictors. J. Abnorm. Psychol. 114, 331–345. https://doi.org/10.1037/0021-843X.114.3.331.
Vermunt, J.K., 2010. Latent class modeling with covariates: two improved three-step approaches. Polit. Anal. 18, 450–469. https://doi.org/10.1093/pan/mpq025.
World Health Organization, 2023. Depression. WHO, Geneva. https://www.who.int/news-room/fact-sheets/detail/depression. (Accessed 1 March 2026).
Yaroslavsky, I., Pettit, J.W., Lewinsohn, P.M., Seeley, J.R., Roberts, R.E., 2013. Heterogeneous trajectories of depressive symptoms: adolescent predictors and adult outcomes. J. Affect. Disord. 148, 391–399. https://doi.org/10.1016/j.jad.2012.06.028.